

Heightened among-individual variation in life history but not morphology is related to developmental temperature in reptiles

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Abstract

Increases in phenotypic variation under extreme (e.g. novel or stressful) environmental conditions are emerging as a crucial process through which evolutionary adaptation can occur. Lack of prior stabilizing selection, as well as potential instability of developmental processes in these environments, may lead to a release of phenotypic variation that can have important evolutionary consequences. Although such patterns have been shown in model study organisms, we know little about the generality of trait variance across environments for non-model organisms. Here, we test whether extreme developmental temperatures increase the phenotypic variation across diverse reptile taxa. We find that the among-individual variation in a key life-history trait (post-hatching growth) increases at extreme cold and hot temperatures. However, variations in two measures of hatchling morphology and in hatchling performance were not related to developmental temperature. Although extreme developmental temperatures may increase the variation in growth, our results suggest that plastic responses to stressful incubation conditions do not generally make more extreme phenotypes available to selection. We discuss the reasons for the general lack of increased variability at extreme incubation temperatures and the implications this has for local adaptation in hatchling morphology and physiology.

KEYWORDS

developmental stress, life history evolution, meta-analysis, phenotypic plasticity, reptiles

1 | INTRODUCTION

Stabilizing selection will eliminate phenotypic variations, resulting in a common, presumably optimal, phenotype or phenotypic reaction norm (de Jong, 1995; Gibson & Wagner, 2000; but see Siegal & Bergman, 2002). This may occur via stabilizing selection favouring developmental interactions that buffer phenotypes against the genetic and environmental perturbations that are unavoidable in nature (i.e. canalization of phenotypic variation; Debat & David, 2001; Waddington, 1942; Wagner et al., 1997).

Novel environments that are rarely encountered, however, are unlikely to have exerted selection on phenotypes, allowing genetic variation, other 'inherited factors' (e.g. parental effects), micro-environmental variation and stochastic developmental errors to become apparent (Moreno, 1994; Rutherford & Lindquist, 1998; Waddington, 1953). As a result, novel, but survivable, conditions may lead to both reduced mean fitness and increased phenotypic variation.

Because phenotypic variation provides the raw material for selection, changes in variance have fundamental evolutionary

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consequences. For example, it may create phenotypic 'noise' that obscures heritable variation slowing responses to selection, or, alternatively, it may reflect a release of cryptic genetic variation that has the potential to contribute to evolvability (Ghalambor et al., 2007; Hansen & Houle, 2008; Levis & Pfennig, 2016; Moczek et al., 2011; Waddington, 1942, 1953; Wagner, 2008; West-Eberhard, 2003). An increasing number of experiments have demonstrated that exposure to novel or extreme environmental conditions can release significant phenotypic variation (Dworkin, 2005; Noble et al., 2019; Queitsch et al., 2002; Rowiński & Rogell, 2017; Rutherford & Lindquist, 1998; Suzuki & Nijhout, 2006; Waddington, 1953). This, in turn, can contribute to the rate and direction of local adaptation during colonization to such environments (Ghalambor et al., 2007; Kelly, 2019; Radersma et al., 2020). However, an increase in variation often arises only in very specific situations, such as within a subset of (sometimes mutant) genotypes (Dworkin, 2005; Rutherford & Lindquist, 1998; Suzuki & Nijhout, 2006), or in response to a short-term stress exposure during a narrow developmental window (Suzuki & Nijhout, 2006; Waddington, 1953). Moreover, studies on model taxa often focus exclusively on those phenotypes that demonstrate diversification or abnormalities, leaving little insight into whether variance in other phenotypes is impacted by stressful conditions (Dworkin, 2005; Queitsch et al., 2002; Rutherford & Lindquist, 1998; Suzuki & Nijhout, 2006; Waddington, 1953).

Although focused research in model taxa has revealed important insight into these areas, it has left us with a crucial blindspot—generality. How common is it that extreme conditions increase phenotypic variation? Across a wide range of taxa, quantitative genetics approaches have revealed that 'stressful' rearing conditions, whether novel or not, typically engender greater genetic and residual variance compared with 'unstressful' conditions (Ledón-Rettig et al., 2010; McGuigan et al., 2011; Noble et al., 2019; Rowiński & Rogell, 2017; Yuan et al., 2016). However, not all species are amenable to quantitative genetic approaches because of long generation times, challenges to conducting controlled breeding experiments or ascertaining parentage in the wild; yet data on phenotypic variance itself are abundant in such taxa. Here, we examine the generality of the variance-environment pattern across a range of non-model taxa using a meta-analytic approach, asking if phenotypic variation increases in response to non-optimal developmental temperatures. Temperature is a crucial feature of an organism's environment, directly impacting metabolic and homeostatic processes (Angilletta, 2009). Temperatures experienced during development can impact population mean values for traits such as size, behaviour and performance (Atkinson, 1994; Colinet et al., 2015; Noble et al., 2018b; Sgrò et al., 2016). Moreover, exposure to stressful temperatures, either during short-term heat shock or a chronic rearing condition, has been shown to increase phenotypic variation in model and non-model animals (Dworkin, 2005; Hoffman & Merilä, 1999; Rowiński & Rogell, 2017; Suzuki & Nijhout, 2006; Waddington, 1953). These effects on phenotypic variation may facilitate local adaptation as populations encounter more extreme thermal environments, a possibility that has become

increasingly recognized but for which the empirical support remains limited (Kelly, 2019).

To understand the generality of increased variation at extreme temperatures, we examined the literature of reptile developmental biology, applying new meta-analytic metrics that are capable of quantitatively comparing changes in phenotypic variance (Nakagawa et al., 2015) to data in the *Reptile Development Database* (Noble et al., 2018a). Our main hypothesis was that prior stabilizing selection has resulted in phenotypic reaction norms that are canalized across a range of intermediate temperatures commonly encountered in wild populations, whereas temperatures beyond the optima lead to a loss of canalization. Thus, we predicted that phenotypic variance within a study would increase at non-optimal temperatures relative to intermediate temperatures (i.e. a U-shaped relationship). However, a suite of other possible hypotheses plausibly exist. For example, an increased variance could be more likely at hot temperatures than cold temperatures due to asymmetries in thermal performance curves leading to hot temperature avoidance across evolutionary time (Angilletta, 2009; Huey & Kingsolver, 1993). Alternatively, variation may be decreased at environmental extremes if poor conditions lead to viability selection or if maximal trait expression varies but depends on optimal developmental conditions (Hoffman & Merilä, 1999; Rowiński & Rogell, 2017). Finally, these latter factors may operate only at the most extreme temperatures and, in concert with the loss of canalization at less-extreme temperatures, create a flat or M-shaped relationship between phenotypic variation and temperature. We considered and compared statistical evidence for all these competing hypotheses.

2 | METHODS

2.1 | Literature collection

We sourced data from the *Reptile Development Database* (www.repdevo.com; Noble et al., 2018a; While et al., 2018). The database is a compilation of available data (1976–2016) on reptile hatchling phenotypes recorded in experiments manipulating egg incubation temperature. Overall, it contains 9,773 trait means from 300 papers and 155 species [from all four reptile Orders—Crocodylia (crocodiles and alligators), Rhychocephalia (tuatara), Squamata (lizards and snakes) and Testudines (turtles)]. Traits in the database are categorized into one of seven categories: incubation duration, survival, development (post-hatching growth and shedding; hereafter 'growth'), performance, physiology, morphology and behaviour. Given the small sample for crocodylians and tuatara ($n = 5$ studies and 22 effects for Crocodylia and $n = 3$ studies and 13 effects for Rhychocephalia), we only used studies on lizards, snakes and turtles. In these two Orders (lizards/snakes and turtles), we examined data on survival (hatching success), growth, morphology and performance as described below. We chose these traits given that they can show evidence of local adaptation, including across thermal environments (e.g. Caley & Schwarzkopf, 2004; Haenel, 2018),

have the largest number of effect sizes and contain measured traits that are generally standardized within the categories. We used only studies that had reported phenotypic outcomes for at least two constant ($\pm 0.5^{\circ}\text{C}$) egg incubation temperatures each with $N \geq 3$ (N = individual animals).

2.2 | Defining 'extreme' temperatures using hatching success

We operationally define which temperatures are extreme or non-optimal as those that show a substantial decrease in hatching success (survival). What qualifies as 'extreme' is not well defined in the literature. As such, we modelled hatching success data and chose two survival temperature thresholds (25% and 50% probability of hatching success—changing the temperature 'extremes'). We used these thresholds to subset the datasets and examine the shape of relationships between phenotypic variance and temperature within each subset. We subsetting the data in this way in case low hatching success substantially contributes to a gain or loss of variation or obscures the link between variation and temperature.

We estimated survival thresholds at the Order level because individual studies each test an idiosyncratic set of temperatures with many having too few temperatures to precisely estimate the full reaction norm. Collectively, however, they cover a large temperature range, thus capturing extremes at both ends of the environmental gradient, as well as the more-optimal centre. Although species and populations undoubtedly vary in which temperatures constitute extreme, there remains a strong taxonomic pattern in the relationship between survival and temperature (for each Order; Du & Shine, 2015; see also Pettersen et al., 2021), meaning related species are relatively similar. Based on this known broad pattern, we operationally viewed temperature extremes as relative to Order-specific survival curves.

Analyses of survival made use of the raw proportion data (proportion hatching) and sample sizes as outcome measures as this has been shown to outperform other effect size measures for dichotomous outcomes, such as the log odds ratio, and simplifies interpretations (Stijnen et al., 2010).

We modelled hatching success using a Bayesian phylogenetic mixed effects model and assumed a binomial error distribution (accounting for overdispersion). We used Bayesian methods as opposed to likelihood-based methods because of convergence issues using the latter and because Bayesian approaches provided greater flexibility in estimating optimal temperatures and their uncertainty. We used the package *MCMCglmm* (Hadfield, 2010) and modelled fixed effects of Order, temperature (linear and quadratic) and their interaction (both linear and quadratic) effects. We also estimated a random intercept and slope (linear and quadratic) at the species level, and accounted for the phylogenetic covariance among taxa in our data. A random effect (intercept) of study was also included in our model. We had 2,100,000 sampling iterations from the posterior

distribution with 100,000 burn-in and a thinning interval of 1000. We ensured that chains were mixing well and that the effective sample size from the chains was >1500 .

To derive the phylogenetic covariance matrix, we obtained a phylogenetic tree of existing taxa using TimeTree.org (Hedges et al., 2006). TimeTree.org generates trees which are based on existing published time-calibrated phylogenies. For species not found within the TimeTree database (after accounting for any conflicts in naming convention), we used existing phylogenetic trees to identify the most closely related species included in the TimeTree database and renamed the species in the TimeTree database to match the species in our data. Thus, we generated a fully resolved phylogenetic tree with branch lengths describing the time between divergence events. The taxa substituted are identified in the online code. We used the *ape* and *phytools* (Paradis & Schliep, 2018; Revell, 2012) packages to trim trees, ensured they were rooted trees without polytomies and generated a phylogenetic covariance matrix for use in our phylogenetic mixed model.

2.3 | Phenotypic variation across thermal environments

To assess canalization of phenotypic variation, we focused our analyses on three large datasets for (1) morphology, (2) growth and (3) performance. We further limited our study to two morphological traits: (1) overall body size, preferably body length (snout-vent length or carapace length, mm), though body mass (g) was used when length was not reported; and (2) tail length. Growth traits were primarily measurements of post-hatching growth rates, but also included days until shedding where the former was not available. Performance traits included measurements of speed and endurance. We focused on these traits because they have been shown to be affected by thermal developmental environments (Hoffman & Merilä, 1999; Noble et al., 2018b; Rowiński & Rogell, 2017), show evidence for local adaptation across populations (Caley & Schwarzkopf, 2004; Haenel, 2018) and exhibit developmental instability in response to extreme temperatures in previous case studies of single species (Brana & Ji, 2000; Kearney & Shine, 2004; Telemeco et al., 2013). Moreover, they are the most commonly reported phenotypes in the database and they are relatively uniform in traits and measurement methods across studies. In a larger sample of the *Reptile Development Database*, mean trait values in these three categories showed modest declines as temperatures decreased to extreme low levels and showed strong declines as temperatures warmed to extreme high levels (Noble et al., 2018b). For traits measured multiple times across an animal's life, we used the earliest record to avoid issues of non-independence (Noble et al., 2017). Although sexual dimorphism is common for many trait types, the vast majority of data came from 'mixed-sex' samples where the sex of each data point was unknown (e.g. 70% of the data from our body size dataset). As such, in cases where the sex of each data

point was known (18% of the data), we took a weighted mean and variance to create a mixed-sex effect size. This made effects more comparable across studies. We also took the weighted mean and variance of effects from experiments that were run at the same temperature across years.

To compare variance across temperatures and studies, we calculated the log coefficient of variance (lnCV; Nakagawa et al., 2015), which can be interpreted as the standard deviation within a group relative to the average trait value across all groups. lnCV and its sampling variance were estimated as follows (based on Nakagawa et al., 2015):

$$\ln CV = \log\left(\frac{s}{\bar{x}}\right) + \frac{1}{2(n-1)},$$

$$s_{\ln CV}^2 = \frac{s^2}{n\bar{x}^2} + \frac{1}{2(n-1)},$$

where s is the standard deviation (SD) of a trait at a given temperature within a study, $\bar{x}$ is the mean of all trait means across temperatures within the study and n is the sample size (number of individuals measured) at the given temperature. We used this effect size for two reasons. First, we require a measure of variance that corrects for inter-study unitary differences, which lnCV does by being relative to trait means. Second, we use the average of means across temperatures within a study to avoid confounding the effects of temperature on the trait mean and the variance. Conventionally, lnCV has been calculated using the mean of the focal group, but in doing so one risks detecting differences in lnCV even though the SD is stable across the groups. We note that lnCV calculated using the mean of the focal group can be helpful in accounting for mean-variance relationships in data (Nakagawa et al., 2015). However, we found mean-variance relationships to be weak or non-existent in most of our datasets after accounting for unitary differences within and across studies (see Supplementary Materials – Figure S1). Lastly, lnCV contains a correction for the biases that sample sizes have on the estimation of variance, allowing us to control for differences in sample size across temperature treatments and papers.

To examine whether variance changes across developmental temperature, we modelled lnCV, weighting effects by their sampling variance, using multi-level meta-analytic (MLMA) and multi-level meta-regression (MLMR) models in the package *metafor* (Viechtbauer, 2010). We first fit a series of models containing null, linear, linear plus quadratic and smooth non-linear (i.e., splines) effects of z-transformed temperature on lnCV. We then selected amongst models using AIC. Collectively, this set of models represents the predicted associations between temperatures and variance derived from the hypotheses in the introduction (U, asymmetric-U, M and flat-shaped). As hypothesized in the introduction, U-shaped relationships in variance may be detectable only within the range of temperatures permissive to viable development if the most extreme temperatures lead to reduced variance (i.e. due to viability selection or reduction in sample size).

TABLE 1 Fixed effects from a generalized linear mixed effect model estimating the probability of hatching as a function of temperature (linear and quadratic) and Order (Squamates and Testudines)

Parameter	Estimate	SE	p-value
Intercept	-38.11	15.34	0.023
T	2.89	0.91	0.01
T^2	-0.05	0.02	0.008
Order(Testudines)	-36.02	22.53	0.094
$T * \text{Order}(\text{Testudines})$	2.35	1.32	0.075
$T^2 * \text{Order}(\text{Testudines})$	-0.04	0.02	0.072

Note: Within-study slopes for temperature are included. Model accounts for possible over/under dispersion of proportion data, study and phylogeny. Estimates are on the log odds scale. Order and interaction parameters estimated are contrasts from Squamates. $R^2_{\text{marginal}} = 56\%$, 95% CI: 40–75%, $R^2_{\text{conditional}} = 65\%$, 95% CI: 52–81%.

Accordingly, we repeated the analyses using only data within the (Order-specific) temperature ranges where hatching success was greater than 25% and 50% (see above). For all models, we calculated both marginal (variance explained by fixed effects only) and conditional (variance explained by both fixed and random effects) R^2 (Nakagawa & Schielzeth, 2010, 2013). Code and data used for all analyses can be found on Figshare at <https://doi.org/10.6084/m9.figshare.4487759>.

3 | RESULTS

3.1 | Impact of thermal environments on hatching success

We obtained 348 estimates from 78 papers of hatching success (lizards/snakes = 194; turtles = 154) for 63 unique species (40 species of lizards/snakes and 23 species of turtles). The temperatures embryos experienced ranged from 6 to 40°C (lizards/snakes = 6–40°C; turtles = 16–37°C). Concordant with a previous study (Du & Shine, 2015), we found strong evidence that the shape of survival curves was a non-linear, quadratic function of temperature, with lizards/snakes and turtles exhibiting different thermal optima and different overall thermal performance curves (Table 1 and Figure 1).

Across species, optimal temperature for survival in turtles was estimated to be 28.96°C (95% CI: 23.82–34.36°C), whereas for lizards/snakes, maximal survival was estimated at an average of 27.45°C (95% CI: 19.54–35.95°C). The difference between the optimal temperatures for the Orders was not statistically significant. Hatching success was predicted to drop below 50% at temperatures less than 24.74°C and above 33.07°C for turtles. The estimated thermal breadth was wider for lizards/snakes, with <50% survival predicted below 22.12°C and above 32.68°C (Figure 1). No hatching success is expected below 13.17°C and 18°C for lizards/snakes and turtles, respectively. At the upper limit, eggs for

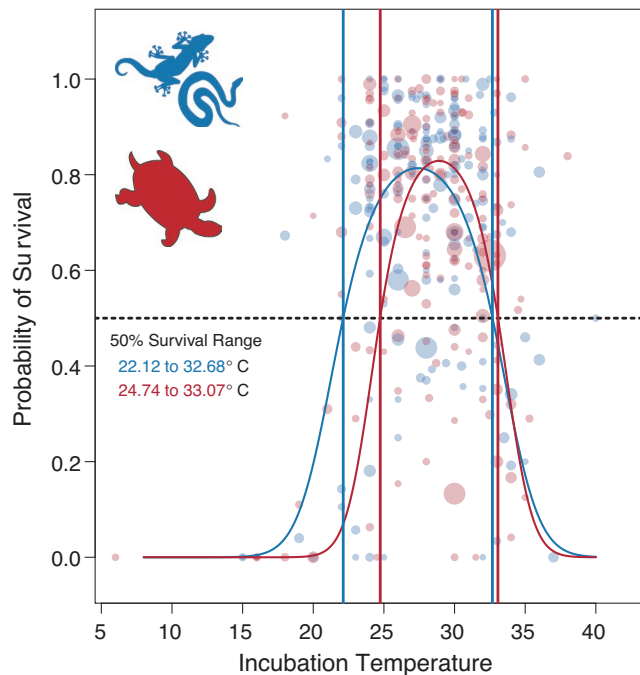


FIGURE 1 Probability of hatching (i.e., survival) in Squamates ('blue') and Testudines ('red') across experimentally varying egg incubation temperatures (6–40°C). Points represent proportion of eggs hatching and point size is proportional to the sample size. Vertical lines represent the 50% cut-off temperatures for Squamates ('blue') and Testudines ('red'). Horizontal dashed line is the 50% survival mark

both groups are predicted to have near 0% survival above approximately 40°C.

3.2 | Does phenotypic variance increase at cold and hot temperatures?

We tested if variability in growth, morphology and performance was impacted by changes in incubation temperature. There was considerable variation in the sample size among trait categories, but fairly consistent ranges of temperatures used across studies (Table 2; growth: 21–32°C, body size: 18–36°C, tail length: 18–34°C and performance: 18–34°C).

For growth, variance increased at low and high temperatures for lizards/snakes while increasing more linearly with temperature for turtles (all data; Figure 2 and Table 2). Non-linear relationships were also present when considering a 25% survival cut-off (Table S2 and S3). However, when restricting the data to only temperatures with at least 50% egg survival, variance increased linearly with increasing temperature for both Orders (Figure S2 and, Table S2 and S3). In contrast, variances in body size, tail length and performance traits did not increase in a linear or non-linear (i.e. quadratic or spline) fashion (Figure 2 and Table 2). The results for these latter traits were the same when using only temperatures with >25% or >50% hatching success (Figure S2, Table S2 and S3).

4 | DISCUSSION

The importance of novel environments for revealing phenotypic variation has re-emerged as a central question in evolutionary theory (Hoffman & Merilä, 1999; Levis & Pfennig, 2016; Moczek et al., 2011; Payne & Wagner, 2019; Rutherford & Lindquist, 1998; Siegal & Bergman, 2002; Wagner et al., 1997; West-Eberhard, 2003). Here, we tested the generality of this pattern across non-model reptile species exposed to different developmental temperatures and found little support for the hypothesis of decreased canalization under 'extreme' conditions. Of four trait categories tested, post-hatching growth was the only trait to demonstrate evidence for higher phenotypic variability at more 'extreme' temperatures compared to intermediate temperatures (in turtles, this was only at warm temperatures). In contrast, morphological and performance traits demonstrated consistent levels of variation across all developmental temperatures.

The mixed results across traits may reflect a biologically meaningful difference between post-hatching growth compared to early hatchling size and performance. Increased variance for life history traits (e.g. growth) compared to morphology has also been shown by Rowiński and Rogell (2017), when focusing on quantitative genetics studies across a broad range of organisms. Life history traits are often considered to be more directly (i.e. causally) linked to fitness and may have, thus, experienced greater prior stabilising selection at common temperatures compared to morphological traits (Rowiński & Rogell, 2017). The variability of life history traits may also reflect that they are composite traits influenced by many morphological, physiological and behavioural traits (Houle, 1992; Mousseau & Roff, 1987; Price & Schluter, 1991). Moreover, the traits herein that lacked a pattern (morphology and performance) were measured soon after hatching and are strongly impacted by maternal investment. For example, morphology at hatching is influenced by egg size, and performance is typically size-dependent (e.g. Sinervo, 1990). In contrast, growth was mostly recorded over ages of more than 1 month. Non-genetic maternal sources of variation may overwhelm other sources of variation at hatching but diminish as animals age and experience different environments. In addition, if maternal effects contribute to developmental buffering equally across all temperatures (e.g. Charmantier & Garant, 2005; Rowiński & Rogell, 2017), early hatchling traits may be less prone to vary as a result of other sources of variation than traits expressed later in life. Study-specific analyses that account for maternal sources of variation (e.g. egg size) may reveal a greater importance of incubation temperature in predicting phenotypic variance in early traits.

In previous studies, increased variance under extreme conditions has been hypothesized to be due to novelty itself (i.e. driven by a lack of prior stabilising selection at these conditions) or due to instability of developmental processes when conditions are stressful (which are not necessarily alternatives; Badyaev, 2005, 2009; Queitsch et al., 2002; Rowiński & Rogell, 2017; Suzuki & Nijhout, 2006). Based on these hypotheses, we predicted for our study that variance would increase at temperatures associated with reduced

TABLE 2 Impact of developmental temperature (T) on trait variance relative to average (lnCV) in lizards/snakes (Order Squamata) and turtles (Order Testudines) for all temperatures. Bold values indicate top supported model and effects that are significant (i.e., $p < 0.05$)

Model	R ²	df	AIC	Model Parameter	Estimate	SE	p
Growth (n = 59, k = 17, s = 7,5)							
ES ~ Order	8% / 56%	4	99.57	Intercept	-1.51	0.15	0.00
ES ~ T * Order	21% / 58%	6	93.03	T	0.13	0.06	0.04
ES ~ T * Order + T ² * Order	26% / 53%	8	89.81	T ²	0.2	0.07	0.01
ES ~ spline(T) * Order	34% / 65%	14	92.53	Order(Testudines)	0.58	0.22	0.01
				T * Order (Testudines)	0.13	0.11	0.27
				T ² * Order (Testudines)	-0.25	0.11	0.02
Performance (n = 132, k = 34, s = 20,9)							
ES ~ Order	2.1% / 8%	4	154.34	Intercept	-1.43	0.1	0.00
ES ~ T + T * Order	2.7% / 9%	6	155.82	Order (Testudines)	0.18	0.2	0.38
ES ~ T * Order + T ² * Order	3.4% / 9%	8	158.05				
ES ~ spline(T) * Order	5.8% / 12%	14	160.30				
Body size (n = 350, k = 83, s = 42,17)							
ES ~ Order	4.6% / 20%	4	331.52	Intercept	-3.03	0.05	0.00
ES ~ T + T * Order	4.7% / 20%	6	335.97	Order (Testudines)	0.24	0.1	0.02
ES ~ T * Order + T ² * Order	4.7% / 20%	8	340.68				
ES ~ spline(T) * Order	5.1% / 20%	14	352.56				
Tail length ^a (n = 160, k = 36, s = 29,0)							
ES ~ Intercept	0.0% / 23%	3	113.60	Intercept	-2.46	0.05	0.00
ES ~ T	0.3% / 23%	4	115.53				
ES ~ T + T ²	0.3% / 24%	5	118.02				
ES ~ spline(T)	1.0% / 25%	8	123.65				

Note: Model selection evaluated the fit of linear, non-linear and non-parametric spline models of the effect of temperature. Estimates are provided for the top AIC-supported model for each trait category. All models include a random effect of study. For each trait category, we present n (number of effect sizes), k (number of studies) and s (number of lizard / snake species, number of turtle species). T is z-transformed to improve model convergence. R^2 (marginal / conditional) for each model are provided.

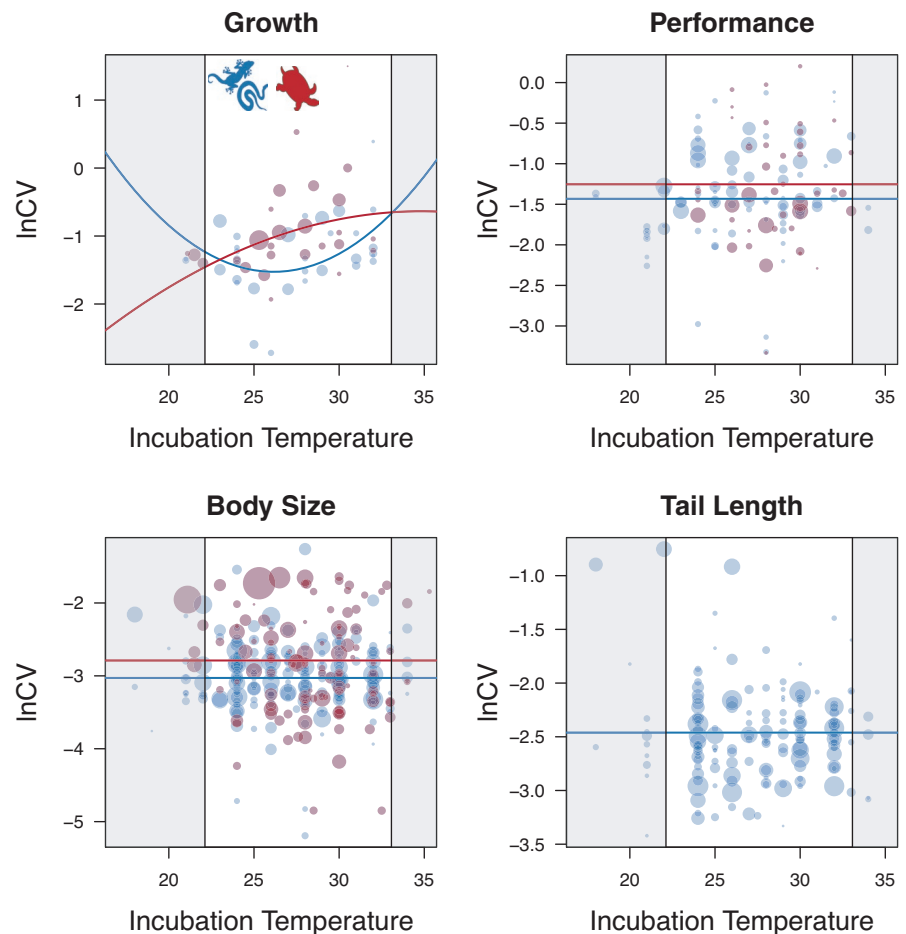
^aDue to a small number of studies on turtles (Testudines) presenting tail length data, only data from lizards/snakes (Squamata) were used in analysis of tail length.

survival, as a rough index of uncommon, potentially stressful temperatures. However, the results do not support a 'canalization across common temperatures' (i.e. novelty) hypothesis for phenotypic variation. In particular, variance in post-hatching growth showed initial increases at temperatures that are not rare among reptiles in the wild (~27–30°C). Mean nest temperatures in reptiles typically fall between 25 and 28°C (excluding Crocodilians and sea turtles; Schwanz et al., 2020), but individual nests are reported above 30°C and below 24°C for many oviparous reptiles (Carter et al., 2019; Telemeco et al., 2013), including 4 species in the dataset (Burger & Zappalorti, 1988; Francis et al., 2019; Löwenborg et al., 2010; Miller & Ligon, 2014). Thus, it is unlikely that increased variance in growth is due to phenotypic variance remaining unseen by natural selection in the wild. These results mirror those of a recent meta-analysis that found the impact of stressful environments on variance had little to do with novelty itself (Rowiński & Rogell, 2017). Whether temperatures in our study are 'stressful' is more difficult to assess. We found that phenotypic variance increased across temperatures that are close to the Order-specific optimal temperatures for hatching success (27.5

and 29°C, herein). However, optimal temperatures are often close to critical maxima (Angilletta, 2009), so temperatures very close to their optima may be developmentally stressful.

Could the weak evidence for temperature-variance patterns in our study be attributed to the lack of a genuinely stressful thermal exposure? Many of the previous studies that have demonstrated conspicuous increases in morphological abnormalities or variance have often employed extreme heat shocks (short exposures to harsh temperature extremes) or manipulated buffering mechanisms such as heat-shock proteins (Queitsch et al., 2002; Rutherford & Lindquist, 1998; Suzuki & Nijhout, 2006; Waddington, 1953). In contrast, the studies in our analyses maintained animals at constant (elevated or depressed) temperatures for prolonged periods of time. Such constant incubation temperatures could be either benign and fully permissive to development or constantly stressful and inviable. In the context of the heat-shock literature, it is tempting to speculate whether increased variance only arises from short-term exposure to stressful conditions but not longer-term exposures. However, cautioning against such conclusions are studies demonstrating that

FIGURE 2 Impact of incubation temperatures on trait variability (lnCV) for Squamates ('blue') and Testudines ('red') for growth, morphological and performance traits. Models are fit to all temperature data available (See Tables S2 and S3 for 50% and 25% survival cut-offs). Point size corresponds to effect size precision (i.e. $1 / se$, where SE is the square root of the effect size sampling variance). Shaded areas of the plots represent the outer bounds of 50% hatching success across both Orders—the lower bound in lizards/snakes (22.12°C) and the upper bound in turtles (33.07°C)



long-term rearing temperatures have been shown to influence phenotypic variance in morphological traits in non-reptilian taxa (e.g. Bublly & Loeschcke, 2002; Guan et al., 2016).

Differences in variance may also be more apparent if environmental extremity could be defined on a species-specific basis. Few published studies on reptiles cover a large enough range of developmental temperatures to provide a robust within-species definition of 'extreme temperature', forcing us (and others) to rely on patterns at the level of Order (see, for example, Du & Ji, 2003; El Mouden et al., 2001; Löwenborg et al., 2010). Although the Order-specific relationships between temperature and survival are strong, phenotypic reaction norms in ectotherms differ across species and populations (Angilletta, 2009; Weber et al., 2012). A more-robust test of increased variance at extreme temperatures would involve species-specific data on hatching success and post-hatching phenotypes across a wide range of developmental temperatures, and preferably for many species (e.g. Löwenborg et al., 2010). Unfortunately, in practice, studies that cover a wide range of temperatures are challenging and thus rare. Due to the nature of the question, designs that can elsewhere be used to limit the total number of animals (e.g. cross-over designs) are not feasible. This very practical issue of scalability then meets the ethical challenges of exposing a large number of vertebrates to temperatures that are anticipated to lead to low survival.

Across reptiles, changes in growth variance are immediately relevant for evolution of natural populations in response to warming environmental conditions. Morphology and physiology commonly show local adaptation and can evolve rapidly, which makes environmental effects on phenotypic distributions worthy of consideration. In reptiles, larger hatchling size often increases survival, foraging and reproductive success in particular environments (Ferguson & Fox, 1984; Langkilde & Shine, 2005; Olsson & Madsen, 2001; Warner & Shine, 2008). Whether the phenotypic variation released by extreme conditions is underpinned by heritable genetic variation is a crucial avenue of future research to enable predictions of population response to environmental change. In *Drosophila* and *Arabidopsis*, the increased variance is associated with genetic variation, leading to increased evolvability (Queitsch et al., 2002; Rutherford & Lindquist, 1998). However, across a wider range of taxa, there is mixed evidence as to whether novel or stressful environments lead to increased genetic variance and heritability (Noble et al., 2019; Rowiński & Rogell, 2017). Much attention in the field of plasticity and evolvability has focused on novel and extreme environments; however, increased phenotypic variation at ecologically relevant conditions in reptile life history traits highlights the potential for environmentally induced phenotypic variation to contribute to evolution under environmental changes.

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PEER REVIEW

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DATA AVAILABILITY STATEMENT

The data that support the findings of this study are openly available in figshare at <http://doi.org/10.6084/m9.figshare.4487759>.

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